

Help Menu

The commands are described under the [Help Menu](#).

Methods of the GanttChart

_Methods of the GanttChart

The *GanttChart* provides:

- The *methods* listed in the table of contents to the left.
- The **Methods of All Objects**.

To view all of the *methods*, *read-only attributes*, and *attributes* of the object, open the window **Show Attributes and Methods**. The figure below illustrates the information using the example of the object *Station*.

The screenshot shows the .Models.Model.Station window with the 'deleteAttr' method selected. The signature is: (NameOfUserDefinedAttribute:string) -> boolean. Red boxes highlight the parameter name and the return type.

Name	Value	Inherited	Watchable	Signature
contentsList	[]			([Contents:table]) -> any
createAttr				(NameOfUserDefinedAttribute:string, DataType:string) -> boolean
createIcon				([IconName:string, Width:integer, Height:integer]) -> integer
deleteAttr				(NameOfUserDefinedAttribute:string) -> boolean

Name of the method: deleteAttr
 Identifier and data type of the parameter you enter: NameOfUserDefinedAttribute:string
 Data type of the return value: boolean

- Select **Show Attributes and Methods** on the context menu of the **Class Library** to show the *methods*, *read-only attributes*, and *attributes* of the selected [Class \[general description\]](#).
- Press the **F8** key or click **Show Attributes and Methods** on the **Home** ribbon tab of the *Frame* into which you inserted an instance to show the *methods*, *read-only attributes*, and *attributes* of the selected [Instance \[general description\]](#).

An example of the **Syntax** line of the individual methods might look like this:

```
<Path>.openDialog([CallOpenControl:boolean:=false]) → boolean
```

- The expression *<Path>* designates the path of the object to which the *method* applies.
- The signature of the method, consisting of the identifier and the data type of the parameter, is listed in parentheses. The expression *(Parameter:string)*, for example, designates a parameter of data type *string*. Instead of a constant value, you can also use a variable of the required type or a *method* that returns the required data type.

Note

Make sure to enter the parentheses for expressions within parentheses (...). Not entering them may lead to unexpected results and open the *Debugger*.

Optional parameters are listed within brackets. The expression *[,Parameter:boolean]*, for example, means that you can, but do not have to enter the *boolean* parameter.

If a parameter has a default value, the signature shows the default value after the parameter, *:= false* in the example above.

If the *method* has a return value, the signature shows its data type after the arrow *->*, *→ boolean* in the example above.

exportChart [SimTalk]

Exports the *GanttChart* designated by <Path> to an HTML file.

Type

Method

Syntax

```
<Path>.exportChart(FileName:string[, TimeScale:real])
```

Parameters

You can specify the following parameters:

- The parameter *FileName* of data type *table* designates the table into which *Plant Simulation* writes the HTML file.
- The optional parameter *TimeScale* of data type *real* designates the time scale. The time scale is shown in pixels per second.
- For the value of 0.5 a Gantt event that has a duration of 60 seconds is displayed by a bar of a length of 30 pixels.

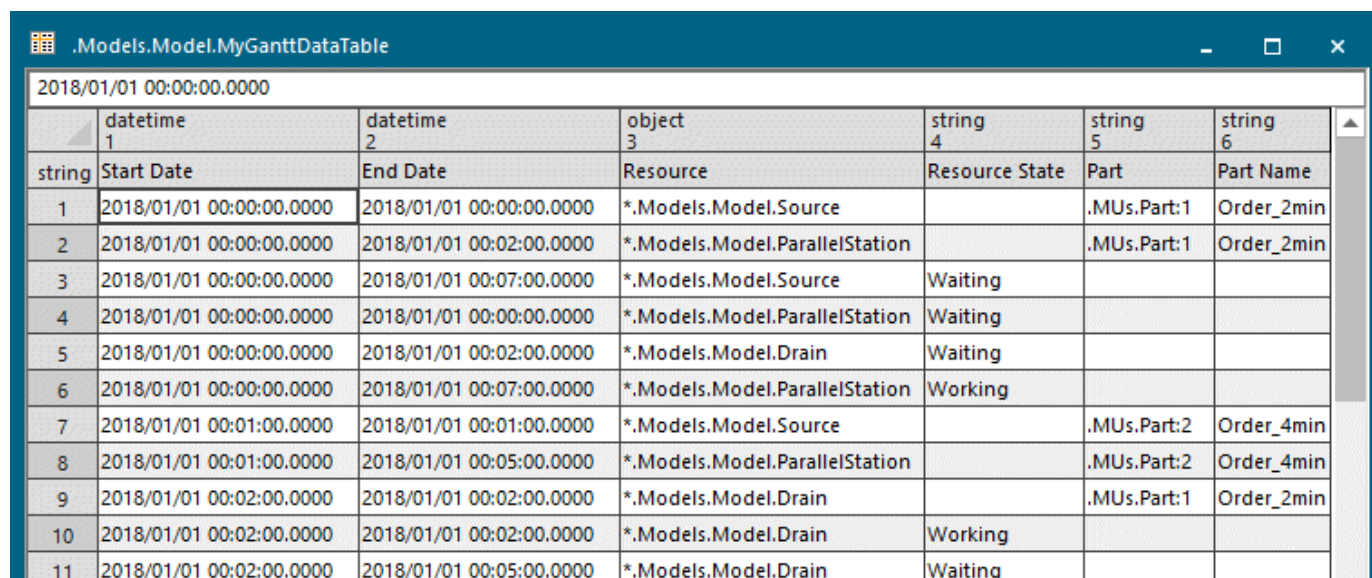
Examples

```
MyGanttChart.exportChart("MyExportedGanttChart.html", 0.5)  
// exports the file to the folder in which the simulation model is located
```

```
MyGanttChart.exportChart("D:\MyModels\MyExportedGanttChart.html")  
// exports the file to the designated folder
```

getData [SimTalk]

Returns the data which the *GanttChart* designated by <Path> shows and writes it to a table.



	datetime 1	datetime 2	object 3	string 4	string 5	string 6
string	Start Date	End Date	Resource	Resource State	Part	Part Name
1	2018/01/01 00:00:00.0000	2018/01/01 00:00:00.0000	*.Models.Model.Source		.MUs.Part:1	Order_2min
2	2018/01/01 00:00:00.0000	2018/01/01 00:02:00.0000	*.Models.Model.ParallelStation		.MUs.Part:1	Order_2min
3	2018/01/01 00:00:00.0000	2018/01/01 00:07:00.0000	*.Models.Model.Source	Waiting		
4	2018/01/01 00:00:00.0000	2018/01/01 00:00:00.0000	*.Models.Model.ParallelStation	Waiting		
5	2018/01/01 00:00:00.0000	2018/01/01 00:02:00.0000	*.Models.Model.Drain	Waiting		
6	2018/01/01 00:00:00.0000	2018/01/01 00:07:00.0000	*.Models.Model.ParallelStation	Working		
7	2018/01/01 00:01:00.0000	2018/01/01 00:01:00.0000	*.Models.Model.Source		.MUs.Part:2	Order_4min
8	2018/01/01 00:01:00.0000	2018/01/01 00:05:00.0000	*.Models.Model.ParallelStation		.MUs.Part:2	Order_4min
9	2018/01/01 00:02:00.0000	2018/01/01 00:02:00.0000	*.Models.Model.Drain		.MUs.Part:1	Order_2min
10	2018/01/01 00:02:00.0000	2018/01/01 00:02:00.0000	*.Models.Model.Drain	Working		
11	2018/01/01 00:02:00.0000	2018/01/01 00:05:00.0000	*.Models.Model.Drain	Waiting		

Type

Method

Syntax

```
<Path>.getData(DataTable:table)
```

Parameter

The parameter *DataTable* of data type *table* designates the table into which *Plant Simulation* writes data that the *GanttChart* shows.

The format of the data table is described under *setData*.

Example

```
MyGanttChart.getData(MyGanttDataTable)
```

SimTalk

[setData \[SimTalk\]](#)

getHTMLCode [SimTalk] - GanttChart

Returns the chart graphic as HTML code in the format SVG of the *GanttChart* designated by <Path>.

Type

Method

Syntax

```
<Path>.getHTMLCode([Caption:string, Scale:real]) → string
```

Parameters

You can specify the following parameters:

- The optional parameter *Caption* of data type *string* designates the desired caption of the graphic. Specify the caption to show the *GanttChart* with this caption.
- The optional parameter *Scale* of data type *real* designates the scaling of the graphic.

Return Value

The return value has the data type *string*.

Example

```
HtmlReport.content := "<<" + GanttChart.getHtmlCode("Test (half size)",  
0.5) + ">>"  
HtmlReport.show
```

See also

[Display a HtmlReport](#)

getLanes [SimTalk]

Returns the lanes which you defined for the *GanttChart* designated by <Path>.

Type

Method

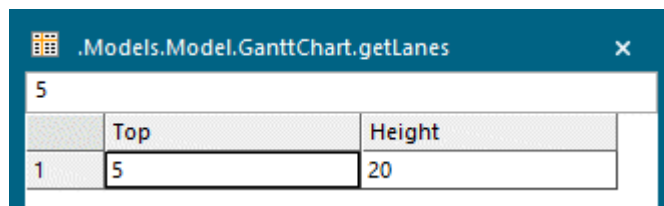
Syntax

```
<Path>.getLanes(TableToWrite:table)
```

Parameter

The parameter *TableToWrite* of data type *table* designates the table into which *Plant Simulation* writes lanes of the *GanttChart*.

A lane is defined by the settings **Top** and **Height**. Both have the data type *integer*.



	Top	Height
1	5	20

Example

```
MyGanttChart.getLanes(MyLanesTable)
```

SimTalk

setLanes [SimTalk]

See also

Edit Lanes

setData [SimTalk]

Sets the data which the *GanttChart* designated by <Path> is going to show.

Remarks

You cannot manually set the data and automatically collect the data simultaneously. When you use the method *setData*, set *CollectData* to `false`.

Type

Method

Syntax

```
<Path>.setData(DataTable:table)
```

Parameter

The parameter *DataTable* of data type *table* designates the table that contains the data which the *GanttChart* is going to show.

The table has eight columns. The figure below shows the data types of the columns. The first three columns have to contain data. Of the second three columns one column at least has to contain data.

.Models.Model.MyGanttDataTable						
2018/01/01 00:00:00.0000						
	datetime 1	datetime 2	object 3	string 4	string 5	string 6
string	Start Date	End Date	Resource	Resource State	Part	Part Name
1	2018/01/01 00:00:00.0000	2018/01/01 00:00:00.0000	*.Models.Model.Source		.MUs.Part:1	Order_2min
2	2018/01/01 00:00:00.0000	2018/01/01 00:02:00.0000	*.Models.Model.ParallelStation		.MUs.Part:1	Order_2min
3	2018/01/01 00:00:00.0000	2018/01/01 00:07:00.0000	*.Models.Model.Source	Waiting		
4	2018/01/01 00:00:00.0000	2018/01/01 00:00:00.0000	*.Models.Model.ParallelStation	Waiting		
5	2018/01/01 00:00:00.0000	2018/01/01 00:02:00.0000	*.Models.Model.Drain	Waiting		
6	2018/01/01 00:00:00.0000	2018/01/01 00:07:00.0000	*.Models.Model.ParallelStation	Working		
7	2018/01/01 00:01:00.0000	2018/01/01 00:01:00.0000	*.Models.Model.Source		.MUs.Part:2	Order_4min
8	2018/01/01 00:01:00.0000	2018/01/01 00:05:00.0000	*.Models.Model.ParallelStation		.MUs.Part:2	Order_4min
9	2018/01/01 00:02:00.0000	2018/01/01 00:02:00.0000	*.Models.Model.Drain		.MUs.Part:1	Order_2min
10	2018/01/01 00:02:00.0000	2018/01/01 00:02:00.0000	*.Models.Model.Drain	Working		
11	2018/01/01 00:02:00.0000	2018/01/01 00:05:00.0000	*.Models.Model.Drain	Waiting		

Each row in the table describes a Gantt event that is visualized as a bar.

- **Start Date** denotes the start date of the event.

- **End Date** denotes the end date of the event.
- **Resource**. Depending on the data of the following three columns **Resource** either denotes the resource that takes the specified state, or the specified part, or both. You can type in the path of the resource in the *Frame* or you can drag the resource on the respective cell and drop it there, or enter free resource text.
- **Resource State** denotes the state in which the **Resource** is. You can enter one of these states: **Working**, **Setting-up**, **Waiting**, **Blocked**, **PoweringUpDown**, **Failed**, **Stopped**, **Paused**, or **Unplanned**. The value does not have to correspond to any existing material flow instance.
- **Part** denotes the part that is processed by the resource. Type in the path of the parts class, a colon, and the number of the part instance. You can, for example, enter `.MUS.Part : 1`. Instead, you can also enter free part text.
- **Part Name** is any string that the *GanttChart* shows as a *Tooltip*.
- **Lane** denotes the lane of the Gantt event.
- **Color** denotes the color of the Gantt event. If you do not enter a color, *Plant Simulation* automatically assigns a color, in **Resource View** according to the part, in **Part View** according to the resource.

Example

```
MyGanttChart.setData(MyGanttData)
```

SimTalk

[getData \[SimTalk\]](#)

[setLanes \[SimTalk\]](#)

See also

[Show Chart \[button\] - GanttChart](#)

[CollectData \[SimTalk\] - GanttChart](#)

setLanes [SimTalk]

Sets the lanes which the *GanttChart* designated by <Path> is to show.

Type

Method

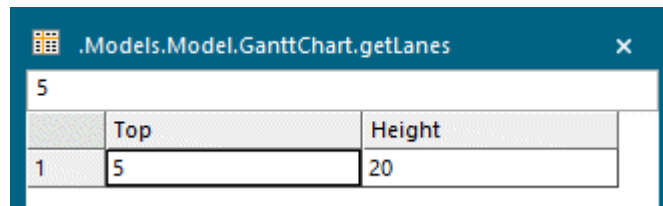
Syntax

```
<Path>.setLanes(TableToRead:table)
```

Parameter

The parameter *TableToRead* of data type *table* designates the table which contains the lanes that the *GanttChart* shows.

A lane is defined by the settings **Top** and **Height**. Both have the data type *integer*.



	Top	Height
1	5	20

Example

```
MyGanttChart.setLanes(MyLanesTable)
MyGanttChart.ShowResourceStates := true
```

SimTalk

[getLanes \[SimTalk\]](#)[Edit Lanes](#)

See also

[Show Chart \[button\] - GanttChart](#)

update [SimTalk] - GanttChart

Refreshes the displayed *GanttChart* designated by <Path> with the current values.

Type

Method

Syntax

```
Path.update
```

Example

```
MyGanttChart.update
```

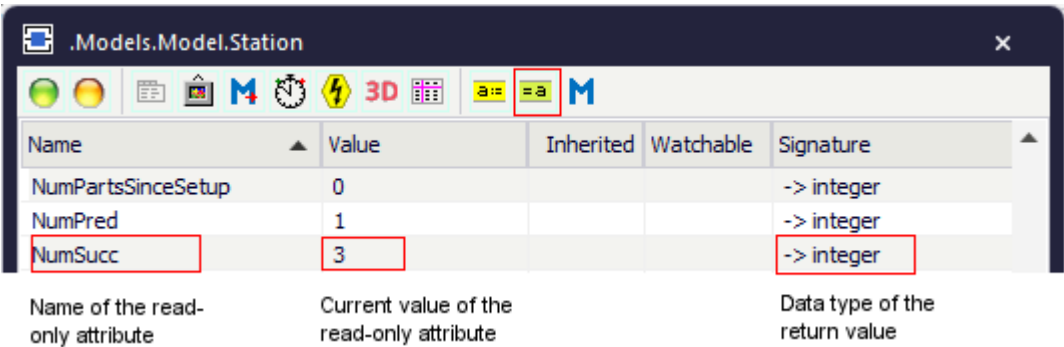
Read-Only Attributes of the GanttChart

Read-Only Attributes of the GanttChart

The *GanttChart* provides the [_Read-Only Attributes of All Objects](#).

You can query the values of the *read-only attributes*, but you cannot set them as *Plant Simulation* computes the value for the point-in-time at which you query it. In most cases a *read-only attribute* corresponds to an unavailable dialog item on one of the tabs of the object, for example on the tab **Statistics**.

To view all of the *methods*, *read-only attributes*, and *attributes* of the object, open the window [Show Attributes and Methods](#). The figure below illustrates the information using the example of the object *Station*.



- Select **Show Attributes and Methods** on the context menu of the **Class Library** to show the *methods*, *read-only attributes*, and *attributes* of the selected [Class \[general description\]](#).